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Candidate Number

BIOLOGY PAPER 1

SECTION B: Question-Answer Book B

This paper must be answered in English

INSTRUCTIONS FOR SECTION B

- (1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7 and 9.
- (2) Refer to the general instructions on the cover of the Question Paper for Section A.
- (3) Answer **ALL** questions.
- (4) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
- (5) Supplementary answer sheets will be supplied on request. Write your candidate number, mark the question number box and stick a barcode label on each sheet, and fasten them with string **INSIDE** this Question-Answer Book.
- (6) Present your answers in paragraphs wherever appropriate.
- (7) The diagrams in this section are **NOT** necessarily drawn to scale.
- (8) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.



SECTION B

Answer **ALL** questions. Write your answers in the spaces provided.

1. For each of the ear parts listed in column 1, select from column 2 one phrase that matches it. Put the appropriate letter in the space provided. (3 marks)

Column 1

Column 2

Ear bones

A. Transmitting vibrations

Eustachian tube

B. Transmitting sound waves

Cochlea

C. Converting sound waves to vibrations

D. Converting vibrations to nerve impulses

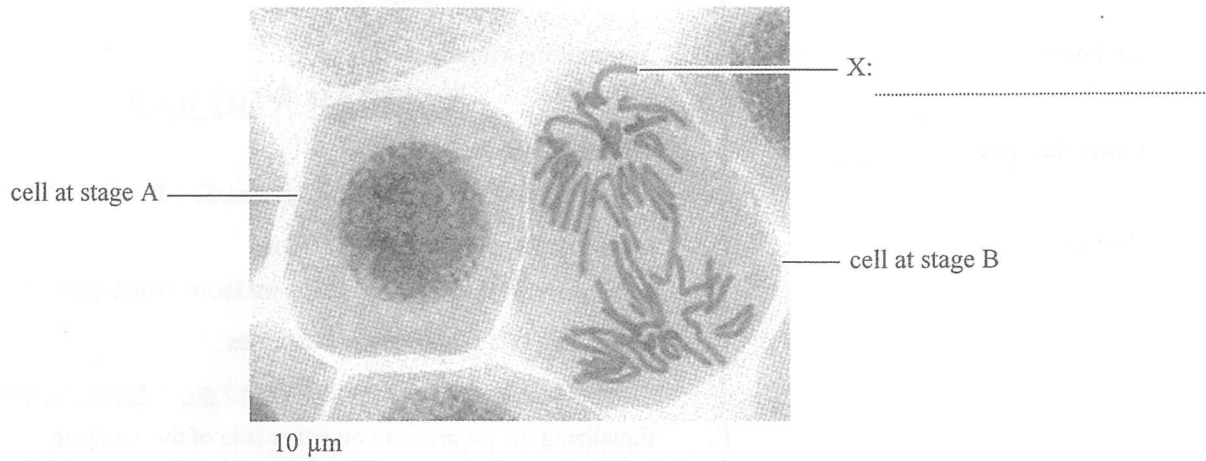
E. Equalizing the air pressure on either side of the ear drum

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2. The photomicrograph below shows the appearance of genetic materials at two different stages of the cell cycle:



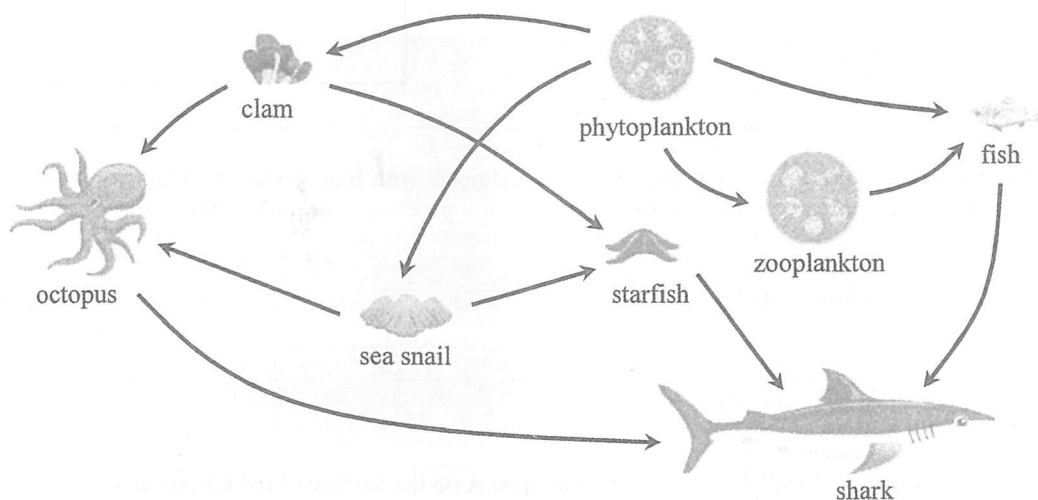
- (a) Label structure X shown in the photomicrograph. (1 mark)
- (b) With reference to the appearance of the genetic materials shown in the photomicrograph, at which stage, A or B, is transcription more likely to take place? Explain your answer. (2 marks)

- (c) In the space provided below, state the cause for the different outcomes of mitosis and meiosis. (2 marks)

	Outcome		Cause
	Mitosis	Meiosis	
Number of daughter cells	2	4	
DNA content in daughter cells	2N	1N	

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3. The diagram below shows the feeding relationships among some organisms in a marine ecosystem:



- (a) Write down the shortest food chain found in the diagram. (1 mark)

- (b) In the space provided below, draw the pyramid of numbers for the food chain in (a). (2 marks)

- (c) Explain the shape of the pyramid of numbers drawn in (b). (3 marks)

- (d) Suggest *two* practical methods that allow you to confirm the feeding relationships among various organisms in this ecosystem. (2 marks)

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4. Roger is found to be suitable for donating blood to recipients with blood types different from his own. However, he cannot receive a blood transfusion from his parents. The blood types of his father and mother are A and B respectively.

(a) What is Roger's blood type? (1 mark)

(b) Given that:

I^A represents the allele for producing antigen A on the surface of red blood cells
 I^B represents the allele for producing antigen B on the surface of red blood cells
i represents the allele that does not lead to the production of any antigens on the surface of red blood cells

(i) Using the above symbols, state Roger's genotype. (1 mark)

(ii) Using the above symbols, state the genotypes of his parents. (2 marks)

Father: _____ Mother: _____

(c) Explain why Roger cannot receive blood transfusions from his parents. (3 marks)

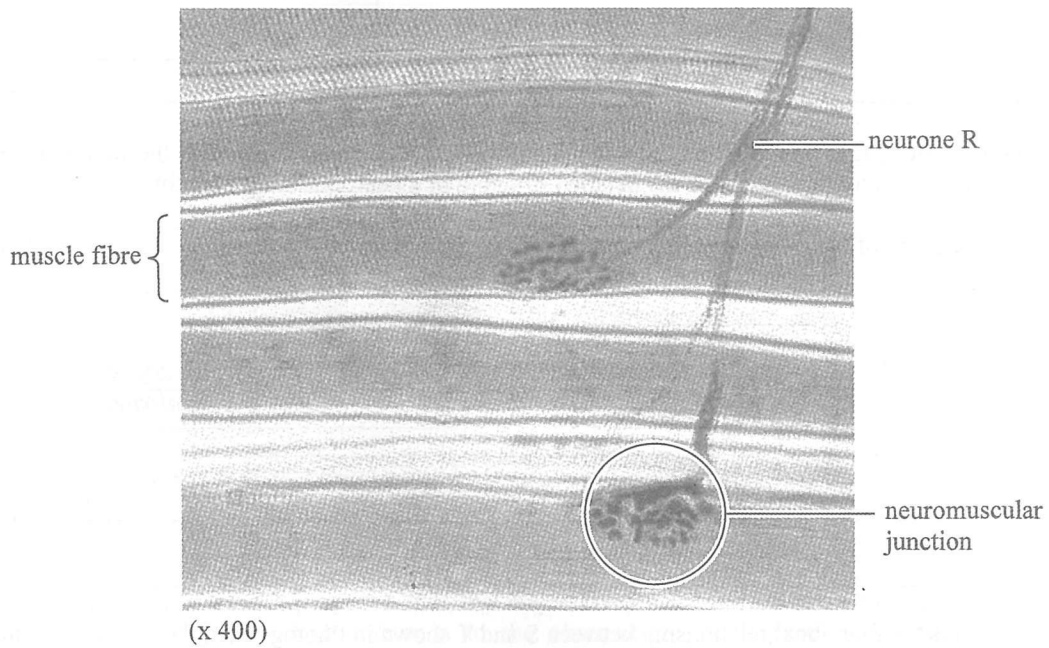
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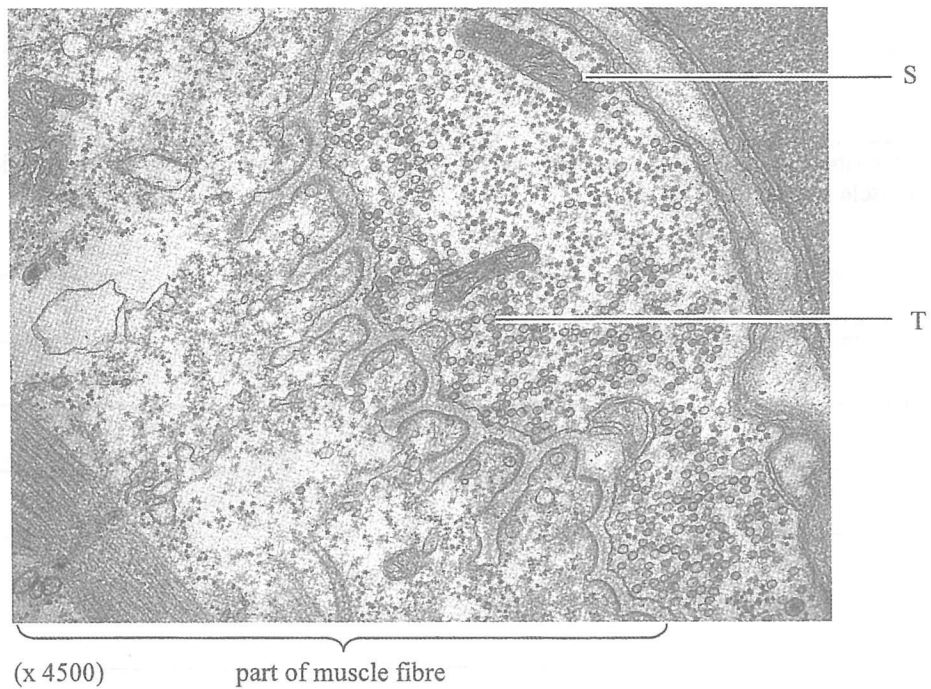
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Photograph P



Photograph Q



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5. On the opposite page, Photograph P shows several muscle fibres associated with the terminal parts of neurone R and Photograph Q shows the detailed structure of a neuromuscular junction.

(a) Which type of neurones does R belong to? Give a reason for your answer. (2 marks)

(b) What is the functional relationship between S and T shown in Photograph Q? (2 marks)

(c) Describe how nerve impulses can be transmitted across the neuromuscular junction leading to muscle contraction. (3 marks)

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6. The table below lists some historical developments about the discovery of the structure of cell membrane:

<i>Year</i>	<i>Scientists</i>	<i>Historical events</i>
1895	Overton	Discovered that lipid-soluble substances could penetrate cells easily
1917	Langmuir	Discovered that the major component of cell membrane exhibited both water-loving and water-hating properties
1925	Gorter & Grendal	Extracted lipids from the cell membrane of red blood cells and spread the lipids in a single layer on a water surface; found that the area of the layer was double the surface area of the cell membrane
1972	Singer & Nicolson	Proposed the Fluid Mosaic Model to explain the structure of cell membranes

- (a) What is the major component noted by Overton and Langmuir? (1 mark)

- (b) Gorter and Grendal proposed that the major component identified in (a) existed as a bilayer (Bilayer Model). With reference to the observation of Langmuir, suggest how this major component is oriented and arranged in the cell membrane. Explain your answer. (3 marks)

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- (c) (i) The Bilayer Model proposed by Gorter and Grendal did not mention another major component of the cell membrane. What is this component? (1 mark)

- (ii) With reference to the Fluid Mosaic Model, briefly describe the orientation of this component in the cell membrane. (2 marks)

- (d) Models are often used by scientists to explain their findings. Complete the following table to elaborate on the aspects of the nature of science involved in the use of scientific models. (2 marks)

<i>Nature of science</i>	<i>Elaboration</i>
Science is evidence based	
	Models are used to simulate an invisible structure or illustrate a theory.

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7. Hydrogen peroxide is a by-product of some metabolic reactions inside our body. If it were allowed to accumulate, it would kill our cells. Fortunately, we have an enzyme called catalase which speeds up the breakdown of hydrogen peroxide into water and oxygen. A student planned to investigate the effect of temperature on the activity of catalase. Below are the steps the student has drafted for his investigation:

- (1) Extract catalase from an animal organ.
- (2) Mix 5 mL 0.1% hydrogen peroxide solution with 1 mL catalase extract.
- (3) Place the mixture in a water bath set at 0°C.
- (4) Measure and record the volume of oxygen gas released in the first 5 minutes.
- (5) Repeat steps (2) to (4) with the water bath set at 20°C, 40°C and 60°C.

- (a) Suggest an animal organ in which catalase is present in great abundance and from which the enzyme can be obtained. Explain why this organ has so much catalase. (2 marks)

- (b) The student has missed out an important step in his drafted procedure. What is it? Explain the importance of this step. (2 marks)

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(c) You are provided with the following apparatus and materials:

measuring cylinder, boiling tube, one-hole stopper, glass tubing, rubber tubing, pipette, ink, water trough, clip

Choose the appropriate apparatus and materials to assemble a set-up for measuring the rate of oxygen production. Draw the set-up in the space below. (3 marks)

Set-up for measuring the rate of oxygen production

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8. In a health check, Lisa was found to have glucose in her urine. She undertook a further check in which she has fasted for 12 hours before a blood sample was taken for examination. The results of the blood tests are shown below:

Tests	Results	Normal range	Units
Blood glucose	8.4	4-6	mmol L ⁻¹
Insulin	0.2	3-32	μU mL ⁻¹
Glucagon	130	20-100	μg L ⁻¹

- (a) State the type of diabetes Lisa is suffering from. (1 mark)

- (b) With reference to the production and actions of the two hormones, account for the results of Lisa's blood tests.

- (i) Insulin (3 marks)

- (ii) Glucagon (3 marks)

- (c) Suggest *two* dietary habits that Lisa should establish. (2 marks)

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9. The photographs below show the appearance of the leaves of a well-watered potted plant at 9 am and 1 pm on a sunny day in summer.

Photograph X (9 am)



Photograph Y (1 pm)



- (a) Briefly describe how the appearance of the leaves shown in photograph X is maintained. (2 marks)

- (b) Suggest an explanation for the appearance of the leaves at 1 pm (Photograph Y). (3 marks)

- (c) With reference to the appearance of the leaves in the two photographs, which one is more effective for photosynthesis? Explain your answer. (4 marks)

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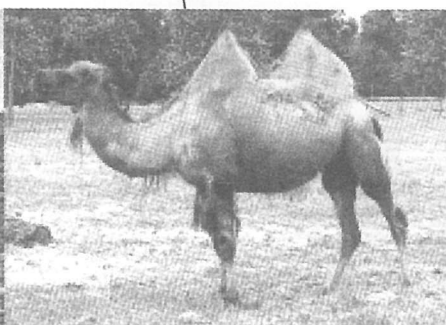
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10. Fossil records suggest that camels in Africa and Asia and llamas in South America evolved from a common ancestor 6 million years ago. The diagram below shows the possible migration routes of the common ancestor at the time before the continents were separated and the locations (shaded areas) where the camels and llamas are found at present:



Camel in Africa



Camel in Asia



Llama in South America

- (a) Based on the information given, draw a diagram to show the evolutionary tree of the three animals. (2 marks)

Evolutionary tree of camels in Africa and Asia and llamas in South America

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- (b) Explain how the common ancestor might have given rise to the two different animal species (camels and llamas) in the above case. (4 marks)

- (c) Suggest another way to establish the evolutionary relationship among the above animals. (1 mark)

- (d) Give *two* limitations of fossil records as evidence for evolution. (2 marks)

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For the following question, candidates are required to present their answer in essay form. Criteria for marking will include relevant content, logical presentation and clarity of expression.

11. Although both arteries and veins are blood vessels, they are very different in their structure. Discuss how their structural differences are related to the different ways of maintaining blood flow inside the blood vessels. (11 marks)

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END OF PAPER

Sources of materials used in this paper will be acknowledged in the *Examination Report and Question Papers* published by the Hong Kong Examinations and Assessment Authority at a later stage.

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